

Logistic Growth, Linearization, and Critical Slowing

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Definition: Half-stability

Let $I \subseteq \mathbb{R}$ be an interval, let $f : I \rightarrow \mathbb{R}$ be continuous, and let $x^* \in \text{int}(I)$ be an equilibrium of $\dot{x} = f(x)$. The equilibrium x^* is **half-stable** if f has the same sign on the two sides of x^* sufficiently near x^* . Precisely, there is a $\delta > 0$ with $[x^* - \delta, x^* + \delta] \subset I$ such that either

$$f(x) > 0 \quad \text{for every } x \in (x^* - \delta, x^*) \cup (x^*, x^* + \delta),$$

or the same statement holds with $f(x) < 0$. Equivalently, the two adjacent phase-line arrows point in the same direction: one toward x^* and one away from x^* .

Observation: A half-stable equilibrium is unstable

For a scalar autonomous system with unique forward solutions, every half-stable equilibrium is Lyapunov unstable because initial conditions on its repelling side leave some fixed neighborhood of the equilibrium.

Definition: Per-capita growth rate

For a population $N(t) > 0$, the **per-capita growth rate** is

$$\frac{\dot{N}}{N}.$$

Definition: Carrying capacity

For a population model

$$\dot{N} = Ng(N), \quad N \geq 0,$$

a number $K > 0$ is a **carrying capacity** if $g(K) = 0$ and there is a $\delta \in (0, K)$ such that

$$g(N) > 0 \quad (K - \delta < N < K), \quad g(N) < 0 \quad (K < N < K + \delta).$$

Idea: The logistic model

Assume that, for $N > 0$, the per-capita growth rate is

$$\frac{\dot{N}}{N} = r \left(1 - \frac{N}{K}\right), \quad r > 0, \quad K > 0.$$

The function $r(1 - N/K)$ extends continuously from $N > 0$ to $N \geq 0$. Using that extension and multiplying by N gives the population model

$$\dot{N} = rN \left(1 - \frac{N}{K}\right), \quad N \geq 0.$$

The factor N makes extinction an equilibrium; the factor $1 - N/K$ models resource limitation.

Definition: Scalar linearization

Let f be differentiable at an equilibrium x^* of $\dot{x} = f(x)$. With

$$u = x - x^*,$$

the **linearization at x^*** is

$$\dot{u} = f'(x^*)u.$$

Definition: Hyperbolic equilibrium

Let f be differentiable at an equilibrium x^* of $\dot{x} = f(x)$. The equilibrium x^* is **hyperbolic** if

$$f'(x^*) \neq 0.$$

Notation: Linearized rate

For an equilibrium x^* of $\dot{x} = f(x)$, the **linearized rate** is

$$\lambda = f'(x^*).$$

Notation: Characteristic time scale

For a hyperbolic equilibrium with linearized rate $\lambda \neq 0$, the **characteristic time scale** is

$$\tau_c = \frac{1}{|\lambda|}.$$

Observation: Critical slowing

As a nonzero linearized rate λ approaches 0, its characteristic time scale satisfies

$$\tau_c = \frac{1}{|\lambda|} \rightarrow \infty.$$

This divergence of the characteristic time scale is called **critical slowing**.

Recall: Local existence and uniqueness

If $I \subseteq \mathbb{R}$ is an open interval and $f \in C^1(I)$, each initial-value problem $\dot{x} = f(x)$, $x(0) = x_0 \in I$, has a unique local solution.

Theorem: Compact-interval continuation criterion

Let $I \subseteq \mathbb{R}$ be open, let $f \in C^1(I)$, and let $x : [0, T_{\max}) \rightarrow I$ be a maximal forward solution of $\dot{x} = f(x)$, where $0 < T_{\max} \leq \infty$. Here **maximal forward** means that the solution has no extension to $[0, T)$ for any $T > T_{\max}$. If there is a compact interval $K_0 \subset I$ such that

$$x(t) \in K_0 \quad (0 \leq t < T_{\max}),$$

then $T_{\max} = \infty$.

Theorem: Scalar linear stability test

Let $I \subseteq \mathbb{R}$ be an open interval, let $f \in C^1(I)$, and let $x^* \in I$ be an equilibrium of $\dot{x} = f(x)$. If $f'(x^*) < 0$, then x^* is locally asymptotically stable. If $f'(x^*) > 0$, then x^* is unstable.

Proof. Since $f(x^*) = 0$,

$$\lim_{x \rightarrow x^*} \frac{f(x)}{x - x^*} = f'(x^*).$$

In either case under consideration, $f'(x^*) \neq 0$. Choose $r > 0$ so that $[x^* - r, x^* + r] \subset I$ and, for $0 < |x - x^*| \leq r$, the quotient has the same sign as $f'(x^*)$.

If $f'(x^*) < 0$, both phase-line arrows point toward x^* . Uniqueness prevents a solution from crossing the equilibrium. Thus a solution starting at $|x_0 - x^*| < r$ stays between x_0 and x^* ; compact-interval continuation makes it defined for every $t \geq 0$. In particular,

$$|x(t) - x^*| \leq |x_0 - x^*|,$$

which proves stability: starting closer keeps the solution at least as close. The solution is monotone and bounded, so it has a limit L . If $f(L) \neq 0$, continuity would eventually give $|\dot{x}(t)| \geq |f(L)|/2 > 0$, which prevents convergence to a finite limit. Hence $f(L) = 0$. The only equilibrium between x_0 and x^* is x^* , so $L = x^*$, proving local asymptotic stability.

If $f'(x^*) > 0$, then $f(x) > 0$ on $(x^*, x^* + r]$. For any $x_0 \in (x^*, x^* + r)$, set

$$m = \min_{x \in [x_0, x^* + r]} f(x) > 0.$$

Until the solution reaches $x^* + r$,

$$\dot{x}(t) \geq m, \quad x(t) \geq x_0 + mt.$$

It must reach that endpoint in finite time: otherwise compact-interval continuation and this inequality contradict $x(t) < x^* + r$ for all $t \geq 0$. Thus initial values arbitrarily close to x^* leave the fixed neighborhood $(x^* - r, x^* + r)$. This is instability. $\square$

Question: What does the logistic model predict?

For $r, K > 0$, analyze

$$\dot{N} = rN \left(1 - \frac{N}{K}\right), \quad N \geq 0.$$

Find and classify the equilibria, compute their characteristic time scales, and solve the equation for $N(0) = N_0 \geq 0$.

Solution. Let

$$f(N) = rN \left(1 - \frac{N}{K}\right).$$

The equilibria are $N^* = 0$ and $N^* = K$. For $N > 0$,

$\frac{N}{f(N)}$		$(0, K)$		(K, ∞)
		+		-

so 0 is unstable from the physical side and K attracts every positive initial population. Moreover,

$$f'(N) = r - \frac{2rN}{K},$$

and therefore

$$f'(0) = r > 0, \quad f'(K) = -r < 0.$$

Thus 0 is hyperbolic and unstable, K is hyperbolic and asymptotically stable, and both characteristic time scales equal $1/r$.

If $N_0 = 0$ or $N_0 = K$, the solution is the corresponding constant solution. For $N_0 > 0$ with $N_0 \neq K$, separation gives

$$\frac{dN}{N(1 - N/K)} = r dt, \quad \frac{1}{N(1 - N/K)} = \frac{1}{N} + \frac{1}{K - N}.$$

Integrating yields

$$\log N - \log |K - N| = rt + C.$$

Imposing $N(0) = N_0$ and solving for N gives

$$\boxed{N(t) = \frac{K}{1 + \left(\frac{K}{N_0} - 1\right) e^{-rt}}}, \quad t \geq 0.$$

The same formula includes the constant solution $N(t) = K$ when $N_0 = K$. For every $N_0 > 0$,

$$\lim_{t \rightarrow \infty} N(t) = K.$$

Question: What if the linearized rate vanishes?

Classify the origin for

$$\dot{x} = -x^3, \quad \dot{x} = x^3, \quad \dot{x} = x^2, \quad \dot{x} = 0.$$

For $\dot{x} = -x^3$, find the solution with initial value x_0 and determine its asymptotic decay.

Solution. Every equation has $f'(0) = 0$, so linearization is inconclusive. Their phase lines give

equation	behavior of $x^* = 0$
$\dot{x} = -x^3$	asymptotically stable
$\dot{x} = x^3$	unstable
$\dot{x} = x^2$	half-stable and Lyapunov unstable
$\dot{x} = 0$	Lyapunov stable but not attracting; every point is an equilibrium

For $\dot{x} = -x^3$, the initial value $x_0 = 0$ gives $x(t) = 0$. If $x_0 \neq 0$, then

$$x^{-3} dx = -dt.$$

Integrating from 0 to t gives

$$-\frac{1}{2x(t)^2} + \frac{1}{2x_0^2} = -t,$$

so

$$x(t) = \frac{x_0}{\sqrt{1 + 2x_0^2 t}}.$$

The maximal interval is

$$\left(-\frac{1}{2x_0^2}, \infty\right).$$

Consequently,

$$x(t) \sim \begin{cases} (2t)^{-1/2}, & x_0 > 0, \\ -(2t)^{-1/2}, & x_0 < 0 \end{cases} \quad (t \rightarrow \infty).$$

The convergence is algebraic rather than exponential.